



End-to-end integrated fully continuous production of recombinant monoclonal antibodies



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1. Introduction

Historically, conversion from batch to continuous processing has resulted in lean, fully automated and agile manufacturing regardless of the nature of an industry (Anderson, 2001; Fletcher, 2010; Laird, 2007; Reay et al., 2013; Tanner, 1998; Thomas, 2008). This conversion has enabled these industries to overcome key business continuity challenges by reducing manufacturing OPEX, CAPEX, and enhancing portability (Utterback, 1994). The Biologics industry is facing similar challenges due to the emergence of competition from biosimilars, cost pressures due to declining industry growth (Kamarck, 2006; Warikoo et al., 2012), and the desire to gain access to emerging markets through standardization and portability. To leverage the lessons learned from other industries, there has been significant interest in the implementation of continuous bioprocessing (Godawat et al., 2012; Konstantinov, 2010; Vogel et al., 2002; Warikoo et al., 2012). This strategic shift is due to the convergence of both the business drivers and advances in downstream technology. Although, the Biologics industry has extensive experience in continuous (perfusion) upstream processing, there

are limited examples of downstream continuous processing due to the lack of enabling disruptive technologies (Godawat et al., 2012; Heeter and Liapis, 1996; Konstantinov and Cooney, 2014a; Lacki and Bryntesson, 2004). However, the past few years have seen maturation of separation technologies such as Simulated Moving Bed (SMB) (Grabski and Mierendorf, 2009), Aqueous Two Phase Systems (ATPS) (Azevedo et al., 2009; Oelmeier et al., 2010; Rosa et al., 2013), single pass TFF (Dizon-Maspat et al., 2011), high capacity membrane absorbers (Orr et al., 2013), and continuous viral inactivation (Caillet-Fauquet et al., 2004; Lorenz et al., 2009). At present, the confluence of these emerging technologies and the business need to innovate is catalyzing the development of continuous bioprocessing.

The Biologics industry is highly regulated to ensure patient safety and, therefore, delivery of consistent product quality is the principal goal of biomanufacturing processes. Inherently, however, the dynamic nature of the Biologics process results in variable product quality as a function of culture days in both perfusion and batch cell culture processes (Pacis et al., 2011; Reid et al., 2010; Robinson et al., 1994). Mohan et al. have specifically shown degradation effects of protease being minimized in continuous-flow cultures as opposed to batch processes (Mohan et al., 1993). Previously, we have demonstrated that a steady state perfusion cell culture process integrated with a continuous capture step can minimize this variability (Warikoo et al., 2012) (Fig. 1); however, in that study, post-capture downstream processing was performed in batch mode. Further benefits of continuous bioprocessing can be realized by developing an end-to-end continuous bioprocess

Abbreviations: Brx, Bioreactor; ATF, Alternating tangential filtration; CAPEX, Capital expenditure; DS, Drug substance; OPEX, Operational expenditure; SMB, Simulating moving bed; ATPS, Aqueous two phase systems; PCC, Periodic counter-current chromatography; mAb, Monoclonal antibody.

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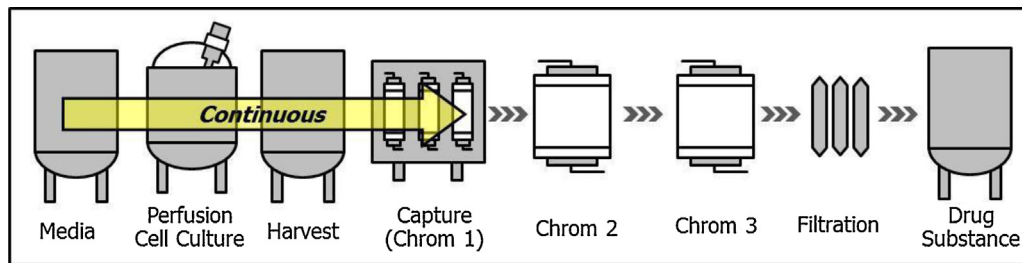


Fig. 1. Continuous upstream + capture, batch downstream.

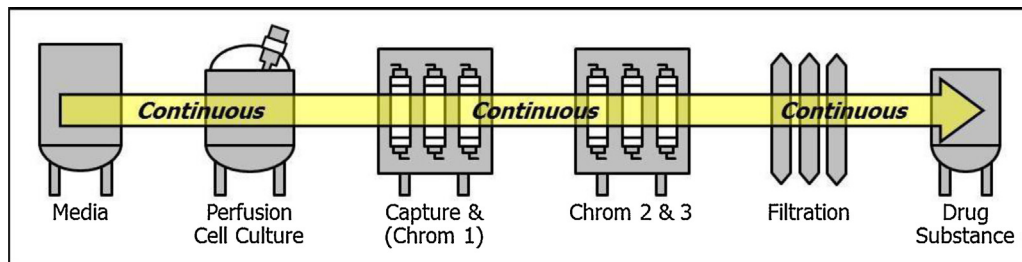


Fig. 2. End-to-end continuous upstream and downstream.

(Fig. 2, (Konstantinov and Cooney, 2014b)). We envision a fully integrated process from bioreactor to formulated drug substance (DS) that is closed, bioburden-free, fully automated with global process control, and operable for long durations. The process should consist of a high cell density perfusion bioreactor connected to a cell retention device, where the cell-free harvest from the bioreactor is continuously captured by a multi-column SMB chromatographic skid. Further, the output from the capture step will be processed downstream by one or more continuous purification steps that are all integrated with one another. The continuous process should also include a concentration and diafiltration step (for example single pass TFF), viral inactivation, viral filtration and sterile filtration. The inputs (raw materials) and outputs (eluates/filtrates) from each unit operation will be constantly monitored using Process Analyti-

cal Technology (PAT) to ensure sufficient traceability and in-process control.

We have used an incremental approach toward the development of fully integrated continuous and closed bioprocessing platform envisioned above. Previously, the continuous flow only included upstream and capture step (Fig. 1). This study demonstrates the feasibility of end-to-end continuous bioprocessing using a mAb as a model protein (Fig. 2). Fully continuous downstream processing can be achieved by any combination of technologies capable of facilitating continuous flow such as SMB in its various modes of operation, membrane adsorbers, precipitation, etc (Hammerschmidt et al., 2015; Sá Gomes and Rodrigues, 2011; Warikoo et al., 2012; Xie et al., 2002). In this study, to develop an integrated and end-to-end continuous process for the production

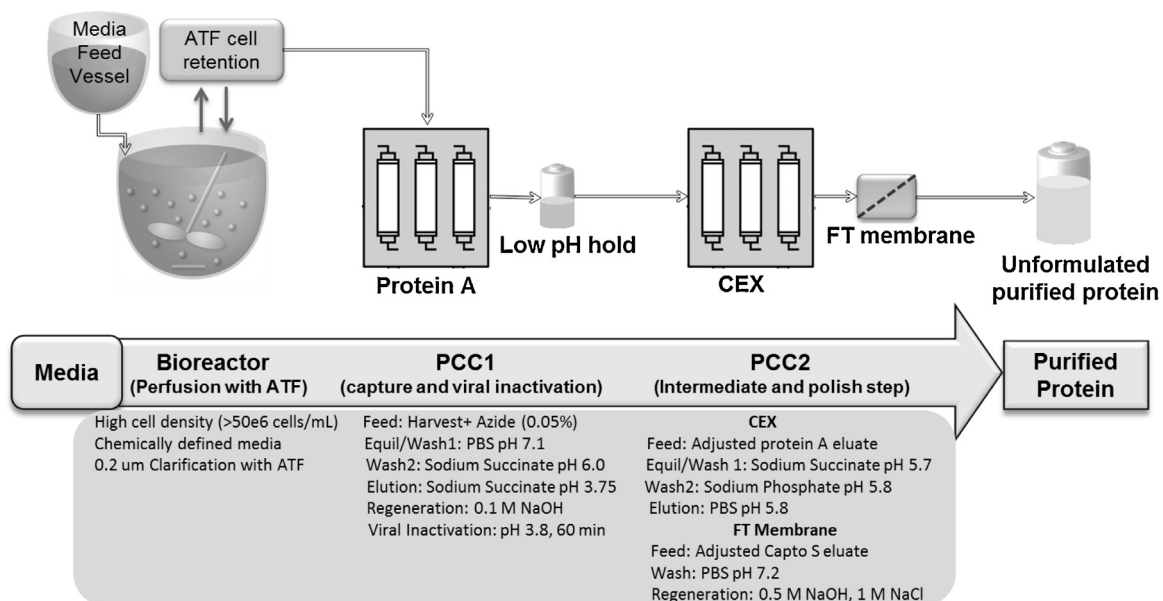


Fig. 3. Architecture of end-to-end continuous bioprocessing. The architecture includes a perfusion bioreactor with ATF as cell retention device for upstream processing. The downstream process included two 4-Column PCC systems. Each PCC system performed two distinct unit operations continuously. The upstream and downstream unit operations were sized and integrated to run in a continuous manner. The details of each unit operation are provided in the shaded region.

of DS, two PCC systems (PCC1 and PCC2) were used. Running in an automated and continuous manner, Brx-PCC1-PCC2 system performed the entire upstream/downstream operation to produce DS. The integrated fully continuous mAb bioprocessing described in this work can serve as a starting point for the development of a universal platform for manufacturing various kinds of therapeutic proteins.

2. Materials and methods

2.1. Integrated upstream and downstream

The fully continuous process included a bioreactor (Brx) with a cell retention system (ATF) and two periodic counter-current chromatography systems (PCC1 and PCC2) for the production of DS (Fig. 3). Each PCC system was programmed to perform multiple unit operations in addition to enabling continuous column operations. The feed harvest was obtained from bench scale 12 L bioreactors as described previously (Warikoo et al., 2012). An in-house proprietary recombinant CHO/DHFR cell line was used in the upstream process. The same CHO clone that met the acceptance criteria for growth and titer stability was used for both batch and continuous cell culture process. Since the engineering of the bench-scale PCC did not allow for closed operation, sodium azide (0.05%) was added to the process stream in-line. This limitation of the small scale system has been successfully addressed by proper design of large scale PCC hardware (Warikoo, 2014).

2.1.1. Periodic counter-current (PCC) chromatographic system

The PCC system used in this study was a custom modified ÄKTA (GE Healthcare, Piscataway, NJ) capable of running up to four columns. The system was equipped with five UV monitors (UV-900), three pumps (P-900), multiple valves (PV-908, SV-903), one pH and conductivity meter (pH/C-900) and a Unicorn-based custom software (GE Healthcare, Piscataway, NJ).

2.1.2. Integration of bioreactor to PCC1 And PCC2

The clarified harvest from the bioreactor/ATF was pumped into a 2 L disposable bag (Hyclone, Logan, UT) serving as a small surge vessel using a peristaltic pump (Masterflex, Cole-Parmer, Vernon Hills, IL). A 0.2 μ m filter (Millipack 40, Millipore, Billerica, MA) was added between the bioreactor and the surge bag as an additional barrier. MabSelect SuRe (GE Healthcare, Piscataway, NJ) chromatographic media in three columns (XK16TM, 1.6 cm $\times$ 6 cm) was used to capture mAb. The eluate from each Protein A column (pH 3.75) was pumped into a 250 mL glass, stirred reservoir with a residence time of one hour to perform viral inactivation (VI). A 0.2 μ m filter (Millipack 100, Millipore, Billerica, MA) was added between PCC1 and PCC2 to act as a mixing device and particulate barrier for downstream columns.

Viral inactivated eluate from PCC1 was further purified on PCC2, using three XK16TM, 1.1 cm $\times$ 7 cm (GE Healthcare) Capto S (GE Healthcare, Piscataway, NJ) columns in bind and elute mode. Finally, Capto S eluate was polished using a 7 ml Sartobind Q membrane (Sartorius Stedim, Goettingen, Germany) connected directly to Capto-S column outlets to yield DS. Additionally, in-line dilution required prior to loading on both the Capto-S and Sartobind Q-membrane was performed using PCC1 and PCC2 pumps, respectively (Fig. 3). Each column operation consisted of equilibration, load, wash, elution and regeneration steps. The Sartobind Q-membrane operation consisted of equilibration, load, wash and regeneration steps. Fig. 3 (shaded area) shows the process parameters for all the unit operations performed using PCC1 and PCC2. The Capto-S column was designed such that the protein eluted in appropriate formulation buffer at required DS concentration. The Capto-S eluate was further polished using Sartobind-Q membrane adsorber

Table 1

Comparison of batch and continuous processing for individual unit operations and entire process. The numbers are normalized to batch processing for the ease of direct comparison.

Parameters	Batch processing	Continuous processing (normalized to batch)
Upstream		
Cycle time*	Days (up to 14 days)	Hours
Equipment utilization	100	110
Productivity (g/L Brx vol/day)	100	> 1100
Downstream		
Cycle time	Days–months	Hours
Resin capacity	100	120
Buffer usage	100	80
Column	100	<5 ($\times$ 3 columns)
Equipment utilization	100	>200
Productivity (g/L resin/day)	100	>600
Whole process		
Automation	Discrete, individual unit ops	Integrated and fully automated
Hold steps	Several (up to 9)	None
Equipment utilization	Low	Very high

* Cycle time refers to residence time of a target molecule in respective unit operation(s)

in a flow through mode, resulting in minimal change in either the protein concentration or buffer formulation, thus eliminating the requirement for terminal UF/DF. However, this approach might not be feasible for all protein drugs, and incorporation of UF/DF in the downstream architecture may be required. This end-to-end bioprocessing architecture does not include a nanofilter in the process train because the pumps and the valves available on the two PCC systems did not allow for addition of another unit operation.

2.1.3. Analytical methods

In-house assays were used for the quantitation of titer, host cell proteins (HCP), aggregation, residual protein A, and potency. Titer was measured using a Protein A column (Applied Biosystems, Carlsbad, CA). Residual protein A and HCP were quantitated by ELISA using antigen and antibodies produced in-house. Aggregation was measured by HPLC-SEC using a TSK-GEL, G3000SWXL, 7.8 mm $\times$ 30 cm, 5 μ m column (TOSOH HAAS, King of Prussia, PA). mAb potency was measured by an in-vitro cell based assay.

2.1.4. Integration of two PCC systems

The predominant operational design goal of the current study was to achieve steady state operation with respect to process flows and mAb quantities held up in the unit operations. To this end, the rate of product input/output of all the unit operations were synchronized based on size, residence time and the frequency of the unit operations. Two PCC's were operated using separate Unicorn based software. The two parameters that dictate the rate of product input/output (R) are the volume of a unit operation (V) and residence time (τ_r) in both columns and membrane absorbers ($V = R \times \tau_r$). Additionally, the frequency (F) of each unit operation depends on the binding capacity (k) and the rate of product input/output ($F = R/k$). In-line product adjustments were performed with appropriate stock solutions at a defined mixing rate using the buffer pumps on the PCC.

3. Results and discussion

The main objective of this study was to demonstrate the feasibility of a fully integrated continuous bioprocessing with a mAb protein over an extended period of time. The process consisted

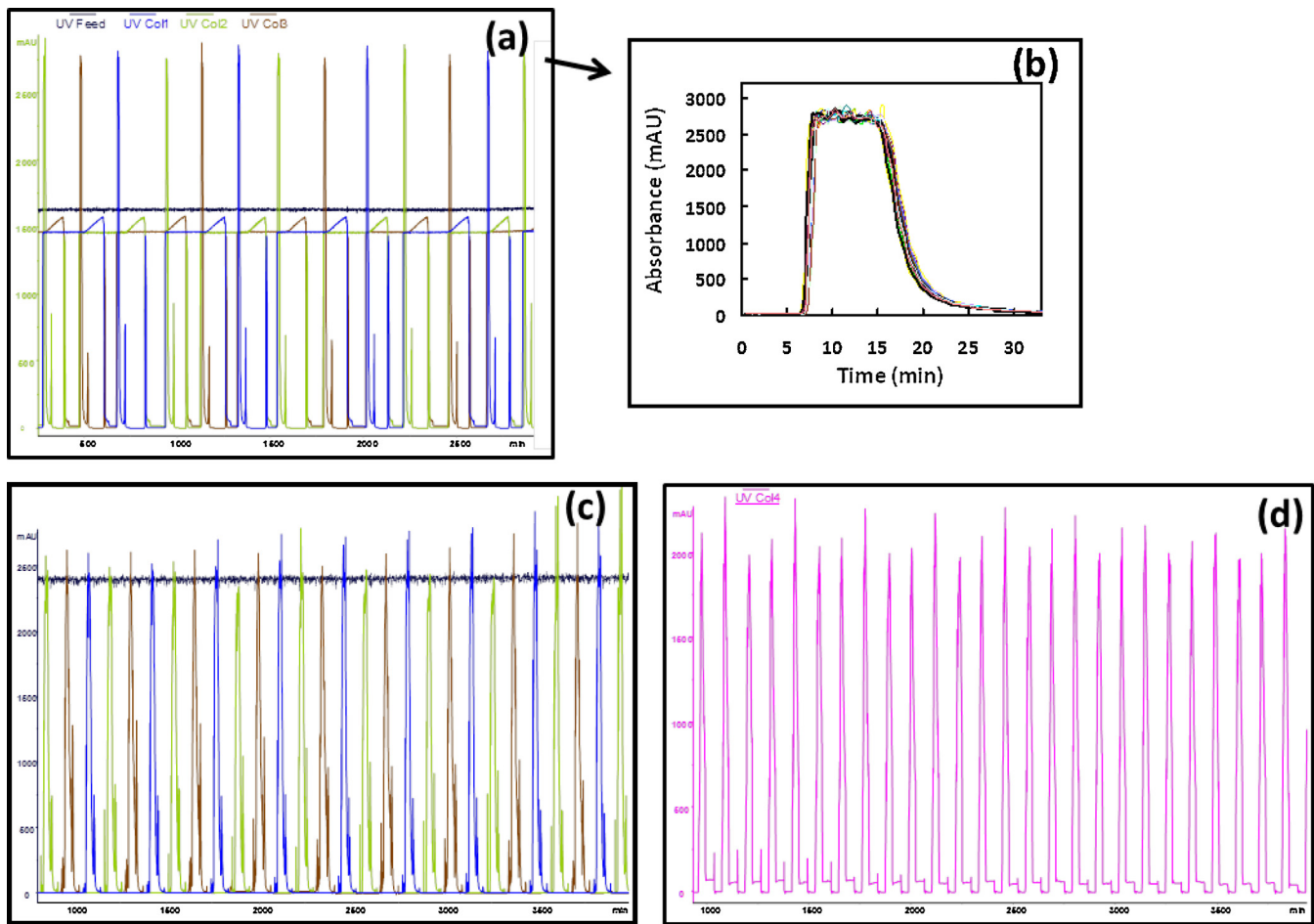


Fig. 4. UV profiles for all downstream unit operations. For clarity, a limited time window is shown instead of the entire chromatogram (a): UV profiles of Protein A capture column. Black – feed UV; blue – column1 outlet UV; brown – column 2 UV outlet; and green – column 3 UV. (b) The zoomed in overlays of all protein A elution peaks for 31 days to show process consistency (c) UV profile representing Capto-S column. Black – feed UV; blue – column1 outlet UV; brown – column 2 UV outlet; and green – column 3 UV. (d) Sartobind Q-membrane filtrate UV profile.

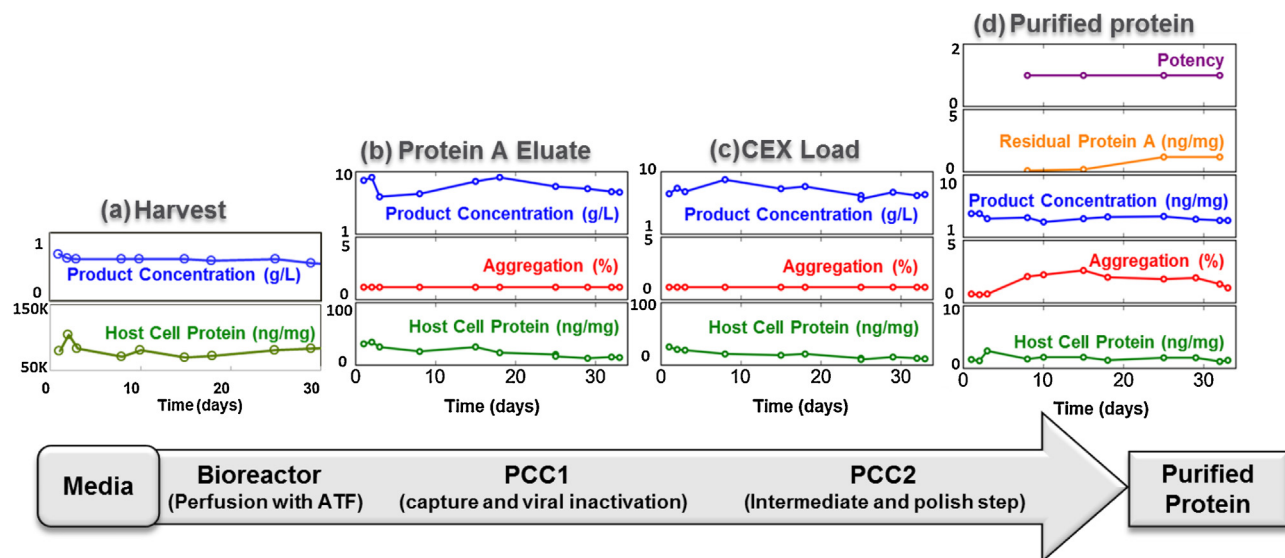


Fig. 5. In-process performance indicators and DS critical quality attributes over 31 days (a) cell culture harvest (b) Protein A eluate (c) cation exchange (CEX) load (d) purified protein. The variability associated with aggregation, and residual protein A was due to the un-optimized Q membrane operation. These trends were comparable between corresponding batch and continuous Q membrane operation (data not shown). All other results above were within the assay variability.

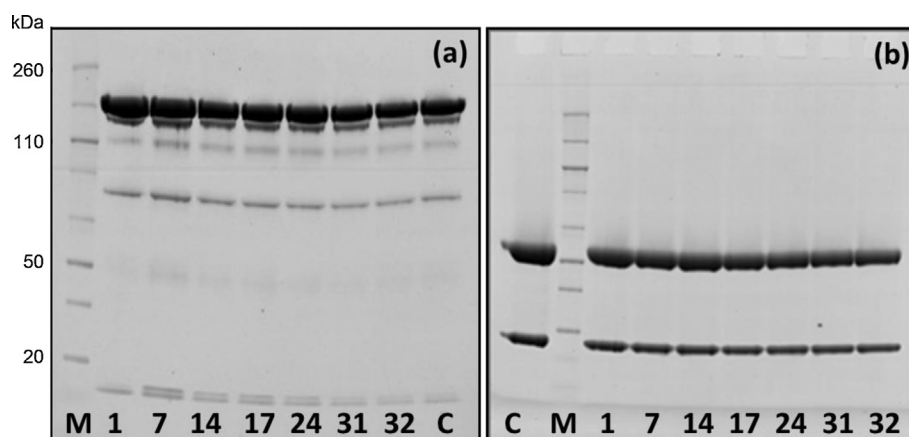


Fig. 6. SDS-PAGE results of purified protein over 31 days of production (a) under non-reducing and (b) reducing conditions. Legend: C is control reference material, M is marker and the number represents day of purification.

of Brx-ATF and downstream operations of capture, viral inactivation and two polishing steps (Fig. 3). The column/membrane sizes and residence times for all unit operations were scaled to achieve continuous purification of the bioreactor harvest to DS based on equations in Section 2.1.4. The continuous operation of the fully automated and integrated system was enabled by balancing the rate of mass and flow across various unit operations. Further, the process intermediates were held only to account for the lack of software/hardware synchronization between the unit operations rather than as a method to balance mass and flow through the process.

The integrated system was operated continuously for 31 days, which corresponds to 31 independent DS batches (a batch is defined as 1 day of processing, 31 Sartobind Q-membrane filtrates). The rate of production of mAb in the bioreactor was ~ 10 g/day and was continuously captured on the Protein A column. The Protein A eluate was generated at ~ 4 h periodic intervals and held for 1 h at low pH for viral inactivation in a stirred tank. The viral inactivated Protein A eluate was in-line adjusted for both pH and conductivity and loaded onto the Capto-S continuously. The Capto-S column eluates were generated every 2 h and instantaneously flowed through the Sartobind Q-membrane absorber as the final polishing step. The DS was generated at a rate of ~ 8 g/day (overall yield $\sim 80\%$). In-process and DS samples were collected and analyzed daily. The cycle time or residence time of the mAb molecule from clarified harvest to unformulated DS was approximately 7 h.

The feed and outlet UV profiles for protein A capture, Capto-S, and Sartobind Q-membrane filtrates were steady state for the entire duration of the integrated Brx-PCC1-PCC2 operation (Fig. 4A–C). The binding capacity for protein A and Capto S was 50 g/L and 81 g/L, respectively. The Sartobind membrane absorber, which was operated in a flow through mode for 30 cycles, was loaded at 90 g/L capacity.

The performance of each unit operation was further monitored by analyzing in-process protein concentration, host cell protein impurity clearance (HCP), and aggregation level (Fig. 5). The process performance was comparable to the batch mode resulting in HCP clearance of 4 logs, removal of residual protein A to 1.0 ppm, and $<5\%$ aggregation. The CQA's showed no time-dependent performance change and were comparable to release specifications of the traditional batch process (Fig. 5). A slight increase in aggregation and residual Protein A results were also observed when the entire downstream process was run in a batch mode (data not shown) indicating that the continuous methodology was not responsible for these results. Further investigation into the anomalous results observed in both the continuous and batch experiments showed

that the root cause was the use of sub-optimal cleaning regimen for the membrane absorber. The DS was also characterized for CQA's such as oligosaccharide profile, potency (data not shown) and purity by reduced and non-reduced SDS-PAGE and the results were comparable between the fully continuous and batch processes (Fig. 6).

Table 1 summarizes the advantages of the Brx-PCC1-PCC2 mAb continuous production in comparison to the existing batch process. Specifically, the volumetric productivity was increased >10 and >6 fold for upstream (g/L Brx vol/day) and downstream (g/L resin/day), respectively. Additionally, chromatography media capacity utilization was increased by 25%, buffer usage was reduced by 20%, and individual column size was reduced at least 20 fold, as compared to batch mode. The combination of a dramatic decrease in cycle time, increased equipment utilization, and increased buffer and resin utilization will result in lower CAPEX and OPEX. Increase in productivity both upstream and downstream, elimination of hold steps, reduction of column size and full system automation meets the design goal of smaller equipment footprint and a simplified overall process (Table 1). Also, as shown in Fig. 5, end-to-end continuous bioprocessing has the advantage of resulting in steady state product quality over prolonged time periods.

4. Implementation of the end-to-end continuous biomanufacturing platform

As shown in Table 1, continuous bioprocessing can significantly and positively impact the operational factors that determine the equipment size and therefore facility footprint. Increased productivity and equipment utilization can result in installation of smaller process trains as compared to the traditional batch processes. In turn, the reduced equipment size can be accommodated in smaller facilities, eliminate scale up requirements and thus streamline and de-risk technical transfers. Also, reduced equipment footprint makes continuous processing amenable to the use of disposable solutions both upstream and downstream.

The potential benefits of end-to-end continuous bioprocessing facilitate the implementation of our vision of a future biomanufacturing facility that meets the lean manufacturing design criteria set forth in the introduction (Fig. 7). The “ballroom” design floor plan utilizes multiple parallel and independent production lines designed as functionally closed “end-to-end” systems that offer multiproduct and multipurpose manufacturing capability. Standardized multiple product lines enable the facility to successfully respond in real time to changing market demands by “numbering up and down”, as opposed to the traditional volumetric scaling

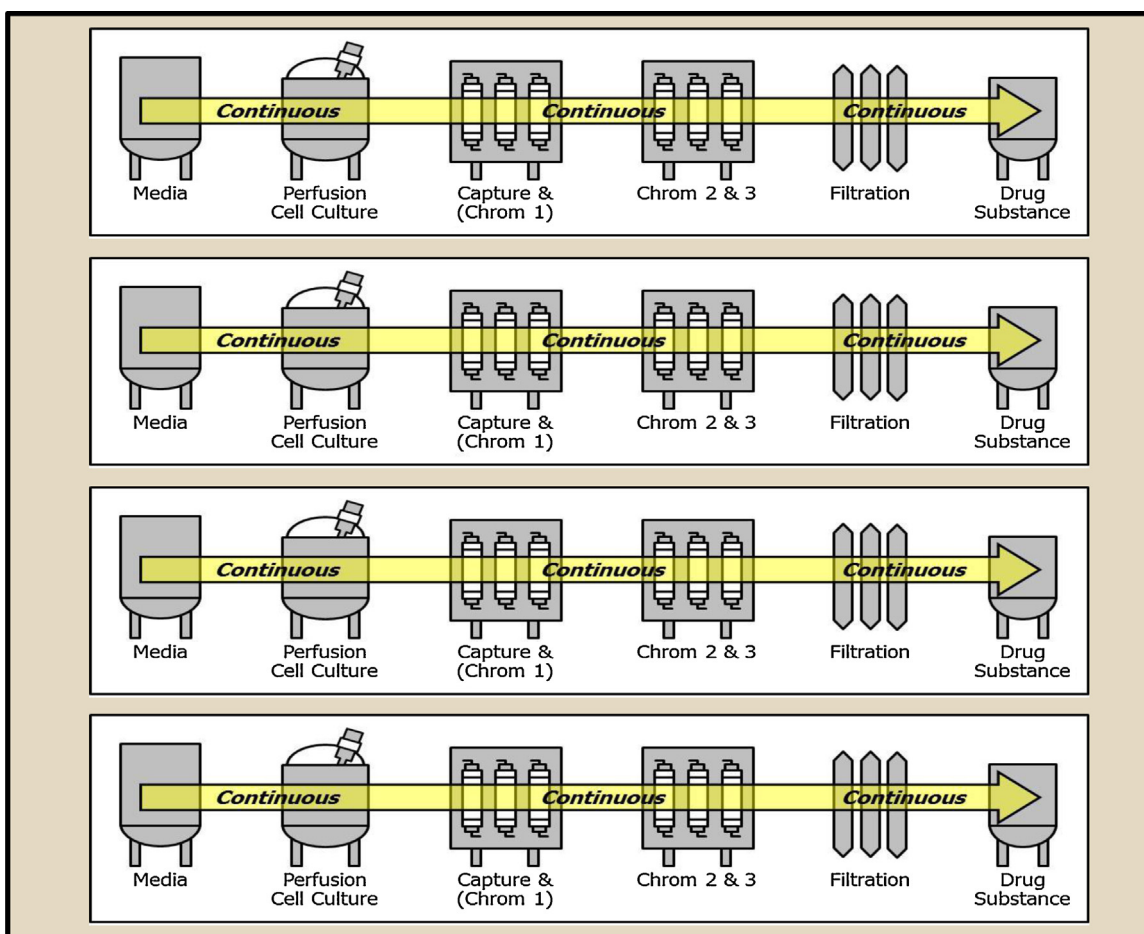


Fig. 7. Ballroom style multiproduct biomanufacturing facility utilizing the integrated continuous bioprocessing platform, implemented as four independent and parallel process trains. The key concepts presented in the text (e.g., closed systems, modularity, small equipment footprint, flexibility, “numbering up” instead of scale up) are an essential part of the facility design.

approach. Due to the closed nature of the process trains, there is less risk of bioburden in the process and the facility can be controlled but non-classified (CNC) thus decreasing both CAPEX and OPEX.

Implementation of disruptive technologies oftentimes requires a shift in organizational mindset and culture, efficient transition from current technologies, and relevant regulatory guidelines (Konstantinov and Cooney, 2014a; Woodcock, 2014). As evidenced by the conversion of continuous to batch by several industries (Anderson, 2001; Reay et al., 2013; Tanner, 1998; Thomas, 2008), such cultural maturity is easier once the technology has been adapted and proven by the early adopters. The development scale studies discussed above are meant to serve as a starting point for implementation of an end-to-end continuous biomanufacturing platform. Additional development efforts are required for a reliable and robust manufacturing process, such as at/in-line monitoring of the product quality using Process Analytical Technology (PAT, (Brower et al., 2014; Read et al., 2010; Tharmalingam et al., 2014)), data management solutions, and validation strategies specifically developed for the continuous process paradigm. Enabling technology such as a manufacturing scale PCC and chromatography columns that are functionally closed and capable of operating bioburden-free for prolonged periods of time developed (Warikoo 2014). However, further efforts are required to establish a robust viral inactivation or removal systems that operate continuously at production scale. The same requirement applies to the UF/DF step of the downstream process. To connect all the unit operations such that they can be operated in an integrated manner requires

development of both a local as well as a global software control strategy.

5. Conclusions

We have used an incremental approach to demonstrate feasibility of an end-to-end continuous bioprocessing platform. Previously, we showed the integration of a perfusion bioreactor to a continuous capture step using both stable (mAb) and relatively less stable molecules (recombinant enzymes) (Godawat et al., 2012; Warikoo et al., 2012). The operational design goals of the current study were (a) process train simplification, (b) uninterrupted and fully automated purification to the DS over an extended period of time, (c) steady state operation with respect to process flows and product quality, and (d) no change with time in the quantities held up in the unit operations. Using a mAb as a model molecule, this study demonstrates the feasibility of end-to-end continuous bioprocessing. The product quality was steady state over the time of operation. The steady state product quality that results from lower residence time can be a marked advantage over batch processes where product stability over culture days can be a significant issue (Joshi et al., 2014; Mohan et al., 1993; Pacis et al., 2011; Reid et al., 2010; Robinson et al., 1994). The consistent product quality over ~30 days also indicates the feasibility of using CHO/DHFR cell line for prolonged duration continuous/perfusion cell culture processes. Also, data shows that end-to-end process results in increasing the process throughput, significantly decreasing the equipment foot-

print, eliminating several non-value added unit operations, such as harvest hold and clarification, eliminating hold steps, and reducing the number of unit operations.

In addition to the enabling technologies that have been used in this study, other novel solutions are also applicable in the development of such end-to-end processes. The technologies amenable to continuous processing include continuous centrifugation, precipitation, crystallization, Aqueous Two Phase Separation (ATPS), multi-zone SMB systems, single pass tangential flow filtration, UVC viral inactivation and many more (Azevedo et al., 2009; Dizon-Maspat et al., 2011; Grabski and Mierendorf, 2009; Hammerschmidt et al., 2014; Lorenz et al., 2009; Rosa et al., 2013).

The economic pressures coupled with the desire to gain access to the emerging markets make process innovation imperative (Gottschalk et al., 2013). The convergence of several factors, such as growing interest in both academia and industry, maturity of the enabling technologies from the vendors, and encouragement from FDA, has driven bioprocessing to a tipping point for process innovation and conversion of bioproduction from batch to continuous processing. An end-to-end bioprocessing platform offers a novel enabling solution to address the inefficiencies of the current non-standardized, large, inflexible, and low throughput facilities. Inherently, continuous flow processes are more conducive to process optimization and control, particularly with respect to steady state product quality. Further, the end-to-end bioprocessing platform is lean, flexible and portable, facilitating implementation in the emerging markets throughout the world.

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